

Talk Like a Structured Graph? Structural Statistics Help Small LLMs Mainly by Stating the Answer

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Abstract

Structural features such as node degree help graph neural networks, but it is unclear whether stating them in text helps a language model that reads a graph as text. We prepend degree, local clustering and random-walk return statistics to GraphQA’s incident encoding and evaluate two Qwen3 models, with and without thinking, on 84,000 paired responses. To separate better graph reading from other explanations, we compare each primer with a length control, measure what a graph-blind solver can answer from the primer alone, and check the graph claims in responses against the edges. Stated degrees give the clearest gains, but mainly by supplying the answer: from degrees alone, the solver answers degree and edge-count queries perfectly, and degrees can hurt a model that already counts accurately. Clustering and return probabilities, which state no answer, have inconsistent effects, and structure-free text also changes answers. Accuracy can also overstate graph use: a small model’s correct cycle answers often cite edges that do not exist. Explicit statistics can help a language model, but mostly by giving answers away, not by improving how it reads the graph.¹

1 Introduction

Graphs have long been the domain of graph neural networks (GNNs), but language models are now also asked to reason over structured data, graphs included. A GNN is routinely helped with precomputed structural features and positional encodings, which supply information that message passing may not readily recover (Dwivedi et al., 2022; Ram-pásek et al., 2022). A language model given a graph as text receives no such help and must extract structure from node and edge descriptions itself, and it does so poorly: on the GraphQA benchmark, Fatemi et al. (2024) find that LLMs struggle with

basic graph tasks. Whether the help a GNN receives would also help a language model, if stated in plain text beside a fixed graph encoding, has to our knowledge not been tested. We therefore ask: *when a black-box language model reads a textual graph, do explicit structural statistics improve execution of graph queries?* Answering it requires care: a degree sentence can itself contain the answer to a degree query, so an accuracy gain need not indicate better use of the edges.

We test this question by placing a primer of degree, local clustering coefficients, or random-walk return probabilities before GraphQA’s incident encoding (Figure 1). We evaluate Qwen3-1.7B and Qwen3-4B, with and without thinking, on paired graph-question prompts across six GraphQA tasks. We also test the combined statistics, a component-count sentence, and a structure-free filler; since added text alone can change a model’s answers (Shi et al., 2023), the filler separates the effect of a primer’s length from that of the statistics it states. A graph-blind solver, which never reads the edge list, gives a lower bound on what the primer makes available; paired comparisons, joint degree and neighbor-set correctness, and checks of response claims against the graph provide further evidence about how answers are obtained.

This controlled evaluation separates information a primer supplies directly from improved use of the incident encoding. Stated degrees help mainly by supplying the answer, while statistics that do not supply it change answers inconsistently.

2 Related Work

Graph representations for LLMs. Work on graphs for language models has largely asked how a graph should be presented. *Talk like a Graph* introduces GraphQA, a set of basic tasks on generated graphs, and compares textual encodings and prompting methods; the encoding, the task and the graph’s structure all change accuracy substan-

¹<https://github.com/GalBrk/GraphTalk>

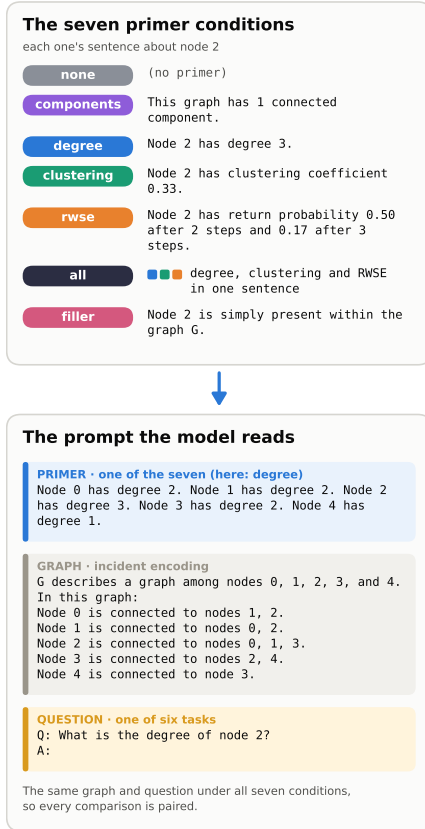


Figure 1: How a primer is used. Top: the seven conditions, each with its sentence about node 2 of a five-node example (the study’s graphs have 40 nodes). Bottom: the prompt the model reads, one primer followed by GraphQA’s incident encoding and the question. The degree and all primers state the answer to this question.

tially (Fatemi et al., 2024). NLGraph poses harder problems, such as shortest paths and maximum flow, in natural language, and finds that advanced prompting helps less as problems grow more complex (Wang et al., 2023). Later work changes the input itself: GraphToken trains an encoder whose outputs a frozen LLM reads as soft graph tokens (Perozzi et al., 2024), and Das et al. (2024) compare describing a graph as text, through its motifs, or as an image. Closest to our question, Huang et al. (2024) ask whether, and why, structural context in a prompt helps an LLM classify the nodes of text-attributed graphs. We hold one encoding, GraphQA’s incident encoding, fixed; since the encoding alone moves accuracy, our findings are specific to it. What we vary instead is a plain-text statement of named statistics, including whether a statistic supplies the queried answer.

Controls and response evidence. Added context, input length, and formatting can change model

answers (Shi et al., 2023; Levy et al., 2024; Sclar et al., 2024). A successful partial-input baseline exposes a shortcut, but an unsuccessful one cannot exclude other shortcuts (Feng et al., 2019). Generated explanations may also misrepresent how an answer was reached (Turpin et al., 2023). Our filler, graph-blind solver and edge-checked claims address each concern.

3 Method

Models and graphs. We evaluate Qwen3-1.7B and Qwen3-4B (Qwen Team, 2025), each with thinking off (P) and on (T); each model–mode pair is an *arm*. We use zero-shot greedy decoding and one response per prompt (budgets in Appendix A). Models and graphs were chosen to leave room for a primer to help: in a pilot on the published GraphQA graphs, Gemma 4 (Gemma Team, 2026) and Qwen3-8B/14B were near ceiling without a primer on most tasks, so any gain would have been too small to detect (Appendix C). We therefore use smaller models on larger, 40-node independent-edge graphs $G(n, p)$ (Gilbert, 1959) at $p \in \{.10, .20, .35, .50\}$, extending node-degree and edge-existence tests to $p \in \{.65, .75, .85\}$. Every condition and arm receives the same 100 graphs per task and density, allowing paired comparisons.

Tasks and primers. We retain GraphQA’s incident encoding and six tasks (Figure 1). Four are the primary measures: node degree, connected nodes, edge count, and edge existence. The other two have constant answers, since every graph has 40 nodes and contains a cycle, so their accuracy cannot show graph reading; Table 1 omits them, and we use cycle-check responses only to assess the support given for a “yes” answer. We score connected nodes primarily by exact match of the predicted and true neighbor sets; set-F1 is a secondary measure that gives partial credit.

Before the encoding, we place either no primer; one sentence per node stating its degree d_v , its local clustering coefficient $c_v = 2t_v/[d_v(d_v - 1)]$, where t_v counts the edges among v ’s neighbors, or its two- and three-step random-walk return probabilities $(P^k)_{vv}$ with $P_{vu} = A_{vu}/d_v$ (a truncated random-walk structural encoding, RWSE; Dwivedi et al., 2022; Rampáček et al., 2022); or a sentence combining all three statistics. We add two controls: a sentence giving the number of connected components, and a filler that names each node without stating a statistic. The filler is similar in length to

Task	Arm	none	degree	clustering	RWSE	all	components	filler
Degree	1.7P	60.50	+7.5 ^f	+3.0	+4.0	+10.2 ^f	+1.8	-0.5
	1.7T	76.25	+8.8 ^f	+1.5 ^f	+2.5 ^f	+12.2 ^f	-1.0	-3.0
	4P	99.25	-6.5 ^f	-0.5	-2.2	-8.0 ^f	+0.2	-0.5
	4T	97.00	+2.5	+1.8	-1.2 ^f	+0.5	+0.2	+2.0
Neighbors	1.7P	72.50	+3.5 ^f	+1.2 ^f	+1.0 ^f	+2.0 ^f	+0.8 ^f	-6.2
	1.7T	78.75	+6.2	+2.0	+0.2	+6.2	+2.5	+2.5
	4P	86.00	-3.0 ^f	+1.0 ^f	+0.0 ^f	-2.8 ^f	+9.2 ^f	-12.5
	4T	95.50	+0.5	-0.8	+0.5	-1.2	+0.5	+1.2
Edge count	1.7P [†]	1.25	+3.0	-0.5	-0.8	+0.8	-1.0	-0.5
	1.7T [†]	15.75	-3.2	-1.2	-3.2	+5.0	-0.8	-1.5
	4P	2.00	+27.2 ^f	-1.2 ^f	-1.5 ^f	+3.8	-1.2 ^f	+1.2
	4T [†]	20.00	+8.5	+24.8	+10.2	+1.8	+24.8	+12.2
Edge exists	1.7P	70.50	+4.0 ^f	+3.8 ^f	+5.0 ^f	+11.5 ^f	-8.2 ^f	-15.8
	1.7T	99.00	-2.5	-1.0	-1.2	-1.8	-0.8	-0.5
	4P	90.50	-4.8	+4.0 ^f	-10.0	-10.0	-2.8 ^f	-7.0
	4T	99.75	+0.0	-0.8	-1.2	-1.0	-0.5	+0.0

Table 1: Primary result on the four tasks whose answers vary (node count and cycle check, whose answers are constant, are in Section 4.5 and Appendix B): correct share without a primer (% of all prompts) and paired change under each primer (points), on the same 400 graphs per row ($p \leq .50$). The degree and all primers supply the degree answer and every term needed to compute edge count. Bold: $q < .05$ against no primer (BH within arm, task and control); ^f: $q < .05$ against filler; [†]: at least 15% truncation on either side in every cell of the row, which therefore carries no significance marks. P/T: plain/thinking.

clustering but not to the other primers (Appendix Table 2). A rule-based graph-blind solver tests what can be answered from the primer without the edge list; it also receives the node and edge counts but never returns them as answers.

Outcomes and inference. Each response is classified as correct, wrong, or truncated. Accuracy uses all prompts as its denominator, so a response that runs out of tokens counts against the condition rather than being dropped; otherwise a primer that made hard prompts run long could look more accurate. The primary effect is the paired change in accuracy, in percentage points, against no primer and against filler on the 400 main-sweep graphs per task (Table 1). We use exact McNemar tests (McNemar, 1947), bootstrap intervals over graphs, and Benjamini–Hochberg correction (Benjamini and Hochberg, 1995) within each arm–task–control family. Cells with at least 15% truncation in either condition are flagged and excluded from pooled summaries. We also report balanced accuracy for edge existence because its positive-answer rate varies with density. A non-significant effect is not evidence of no effect (detectable sizes in Appendix A).

Evidence of graph use. Because degree and neighbor-set questions target the same node, their joint correctness tests whether a degree gain accompanies correct identification of that node’s neighbors. Rule-based classifiers label each response. A

lookup answer does not restate the queried node’s incident line; an *enumeration* restates it and lists at least five neighbors on separate lines; a *conflict* response notes that its own count disagrees with the printed degree. For correct, finished cycle-check answers, we extract every closed walk the response writes out and check each step against the edge list: a *real cycle* passes through three or more distinct nodes with every step an edge, an *invented cycle* includes a nonexistent edge, and a closed walk of two nodes or one that retraces its steps is *not a cycle*; the remaining answers name no cycle. Among those, we also identify valid arguments from edge count: a 40-node graph with at least 40 edges must contain a cycle.

Follow-up experiments. Beyond the main sweep, three node-degree experiments with Qwen3-1.7B retest the small clustering gain that a 40-cell screen turned up: 400 graphs per density at all seven densities, freshly generated graphs, and graphs of 20–160 nodes at fixed mean degrees 8 and 16, which also include Qwen3-8B (Appendix C).

4 Results

4.1 What the primers make available

From stated degrees alone, the graph-blind solver recovers node degree directly and edge count by halving the degree sum, both with 100% accuracy. Without degrees, the rule-based solver reaches at

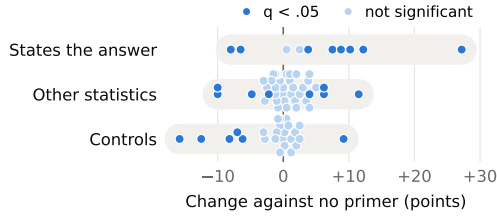


Figure 2: **Large gains come mainly from primers that state the answer.** Each dot is one cell of Table 1: the change in one arm’s accuracy on one task when a primer is added (points; right of zero is a gain; cells with at least 15% truncation omitted). Rows group cells by what the primer tells the model: *states the answer* (degree and all on node degree and edge count), *other statistics* (every other primer–task pair) and *controls* (components and filler). Filled: BH $q < .05$ against no primer. Other statistics move accuracy both ways, about as widely as the controls, and every significant filler effect is a loss.

most 21.0% on degree and 10.4% on edge count, compared with 14.4% and 3.0% without a primer (Appendix Table 2). The solver’s score measures what a primer makes available, not what a model will achieve: every encoding names all 40 nodes in its first line, yet plain 1.7B answers node count correctly on only 31.50% of prompts without a primer (Appendix Table 3), and stated degrees leave every model short of the solver (Section 4.2). The models’ largest gains likewise come where the primer states the answer (Figure 2).

4.2 Stated degrees can help or disrupt counting

The degree primer raises node-degree accuracy by 7.5 points for plain 1.7B and 8.8 for thinking 1.7B, but lowers it by 6.5 for plain 4B, whose unaided accuracy is 99.25%; for thinking 4B, at 97.00% unaided, its +2.5 points are not significant (Table 1). Its effect depends on the cost of counting from the incident list. Within plain 4B, it harms queries about the lowest-degree nodes and helps those about the highest-degree nodes (−11.8 and +10.6 points across seven densities); at $p = .85$, where unaided accuracy falls to 33.0%, it gains 27.0 points (Appendix B). Stated degrees also supply every term needed for edge count, raising plain 4B’s main-sweep accuracy from 2.00% to 29.25%, far below the solver’s 100%.

Stated answers are not always used. Across seven densities, plain 1.7B is correct on 43% of degree prompts under the degree primer, against 39% without it; none of the answers the primer turns from wrong to correct at $p = .20$ and .35

cites the stated value. Plain 4B, by contrast, appears to look the answer up: at $p = .50$, 62% of its responses answer directly and only 8% list the neighbors; across densities, its wrong lookups match the degree printed one sentence away more often than chance (Appendix B); its losses under the degree primer concentrate where the response pattern changes.

4.3 Primers also change answers they do not supply

Filler alone flips up to 18.6% of an arm’s outcomes against no primer, showing that added text can matter without adding a statistic (Appendix B). Clustering and RWSE mostly have smaller effects than degree (Figure 2) and can oppose each other: on plain 4B’s edge existence, clustering raises balanced accuracy by 3.0 points while RWSE lowers it by 6.5 (Appendix Table 6). Clustering’s 3.0-point gain on plain 1.7B degree queries is not significant in Table 1 but replicates on unscreened graphs and a fresh seed; it is not detected at $p \geq .65$ or in Qwen3-8B (Appendix C). Combining the statistics can reverse their separate effects: on plain 4B’s dense degree queries, degree, clustering, and RWSE each raise accuracy (9.3, 11.3, and 7.7 points), but the combined primer lowers it by 21.3 on the same graphs (Appendix B).

The largest unflagged gain without degrees comes from the component sentence, which is identical across graphs at $p \geq .20$: it raises plain 4B’s exact neighbor-set accuracy by 9.2 points, mainly by making responses restate the queried node’s incident line (Appendix B).

4.4 Primers in thinking mode

Thinking mode leaves primers little to add, and primers rarely substitute for it. Thinking 4B is near ceiling on degree, neighbor sets, and edge existence; every primer changes these accuracies by at most 2.5 points, and even across all seven densities it answers 98% of node-degree prompts correctly under the degree primer, against 97% without. Thinking 1.7B, which counts less reliably, gains from stated degrees most where counting is hardest: +16.0 points across seven densities and +25.7 at $p \geq .65$. It does not simply copy the value: it still lists the neighbors in 90% of responses at $p \geq .35$ and checks its count against the printed one, which raises its node-degree truncation rate from 1.1% to 7.6%; when the two conflict, it ends with the stated value in 66% of finished cases.

Nor do primers close the gap between the modes: where thinking outperforms plain mode, they rarely close it, and for plain 4B several widen it (Appendix Table 8). The exceptions include the combined primer, which cuts plain 1.7B’s node-degree gap from 15.8 to 5.5 points, and the component sentence, which brings plain 4B’s neighbor-set exact match within the ± 3 -point equivalence margin of thinking 4B.

4.5 Is a correct answer backed by the graph?

Accuracy credits an answer however it was reached, so two checks ask whether correct answers rest on the edges. First, a model that reads the queried node’s incident line should get both its degree and its neighbor set right. Clustering and RWSE do not significantly improve the joint correctness of degree and neighbor-set answers for the same node in any arm (Appendix B). The degree primer raises this measure by 8.5 points for plain 1.7B but lowers it by 9.2 for plain 4B; the component sentence raises plain 4B’s by 9.2.

Second, since cycle-check accuracy cannot show cycle detection (Section 3), we examine the support in correct, finished “yes” responses (Appendix Table 10). Without a primer, 43.5% of plain 1.7B’s correct, finished responses cite an invented cycle containing a nonexistent edge. Counting over all prompts, valid cited cycles or edge-count arguments support 34.2% and 58.5% for plain 1.7B and plain 4B, respectively. The degree primer shifts plain-model responses toward valid edge-count arguments, which need no named cycle; plain 1.7B’s share of prompts with valid support rises to 50.2%, although its cycle-check accuracy falls by 17.5 points as truncation rises by 8.2. A correct “yes” can thus rest on a valid proof, an invented cycle, or no cycle at all.

5 Discussion

Structural statistics serve a different role here than in graph models, where random-walk encodings can distinguish structures that some message-passing networks cannot represent (Xu et al., 2019; Dwivedi et al., 2022). In our setting, the incident encoding already contains every edge, but the language model must find the relevant edges and reason over them. Stated degree can bypass that computation for degree and edge-count queries, but it also introduces a lookup that can fail when the model selects the wrong node’s sentence.

Clustering and RWSE, by contrast, state no answer, and their mixed effects, like those of filler and the component sentence, are consistent with primers changing how models use the prompt. Independent-edge graphs give these statistics little to convey beyond degree and density, so their behavior on graphs rich in triangles or return patterns remains open.

5.1 Future work

The most direct extensions scale what we ran. Larger graphs and harder tasks, such as shortest paths (Wang et al., 2023), would restore headroom for the larger models the pilot left near ceiling, and other model families would test whether the effects are specific to Qwen3. Because the encoding alone changes accuracy (Fatemi et al., 2024), primers should also be crossed with GraphQA’s other encodings, and more graphs per cell would detect the small side-statistic effects this design cannot. Acyclic graphs, and families that vary triangles and return patterns at matched size and density, would give cycle check two answers and let clustering and RWSE carry information.

A second direction replaces the fixed primer with interaction. An agent that queries the graph iteratively, for instance by requesting one node’s neighbors at a time, would test whether stepwise access helps where a single long prompt does not. Offering the statistics as a tool the model can call would show whether a model knows when a statistic is worth computing, and when a primer is the cheapest way to supply it.

6 Conclusion

Do explicit structural statistics help a language model execute queries over a textual graph? In our tests, not consistently. Stated degrees gave the clearest gains, but mainly by supplying the answer: they helped where a model counted unreliably yet could use the printed value, and hurt plain 4B, which already counted accurately. Clustering and RWSE, which state no answer, had task- and density-dependent effects and never significantly improved joint degree and neighbor-set correctness, while a nearly constant component sentence produced the largest gain without stated degrees, so primers also change how a model uses the prompt. Judging a primer thus needs a graph-blind solver, a length control and graph-checked answers: a correct cycle answer can cite a nonexistent edge.

Limitations

- **Scope.** We test two Qwen3 models (Qwen3-8B in one follow-up), one incident encoding, fixed prompts, and mostly 40-node independent-edge graphs.
- **Cycle check.** All sampled graphs contain cycles, so cycle analysis evaluates support for “yes” answers, not cycle discrimination.
- **Controls.** The filler only roughly matches clustering’s length, and component count supplies structural information. We did not test an instruction-only control; the queried node’s list position and identifier length are confounded with each other.
- **Graph-blind solver.** A solver that fails to recover an answer cannot exclude other clues in the primer.
- **Truncation.** Fixed budgets cause substantial truncation in some thinking-mode cells, and reruns can differ.
- **Response categories.** They describe behavior, not internal computation: only conflict classifiers were checked against blind hand labels; the others received supervised spot checks.

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A Method Details

Primer statistics. For the undirected adjacency matrix A , let $d_v = \sum_u A_{vu}$ and $P_{vu} = A_{vu}/d_v$ when $d_v > 0$ (a zero row otherwise). If t_v is the number of edges among v ’s neighbors, local clustering is $c_v = 2t_v/[d_v(d_v - 1)]$, set to zero for $d_v < 2$. The two return features are $r_v^{(k)} = (P^k)_{vv}$ for $k \in \{2, 3\}$. Floats are printed to two decimals, with one sentence per node in node order (Table 2).

Feature choice. Degree and random-walk encodings are used in graph Transformers (Rampášek et al., 2022); clustering records edges among a node’s neighbors. We use return steps 2 and 3: step 1 is zero in a simple graph, while step 3 reflects triangles. We omit Laplacian coordinates because their signs and degenerate eigenspace bases require a convention (Dwivedi et al., 2022). Component count is a global control, though it is informative in principle: a graph with n nodes, m edges and c components is acyclic exactly when $m = n - c$.

Models, budgets and inference. We run the Qwen3 checkpoints with transformers. The output limit is 8,192 tokens, except 2,048 for plain arms at $p \geq .65$; the shorter limit costs little, because, pooled over primers, no arm and task in the high-density extension truncates more than 2.5%. Because Table 1 compares 400 paired graphs, its non-significant cells rule out only moderate gains: the median detectable gain is 5.7 points in the 1.7B arms and 3.2 in the 4B arms, and 18 of thinking 4B’s 28 null comparisons sit too close to ceiling for any gain to be detectable. A single 100-graph density cell needs 7.4 points, and side statistics (primers that do not state the queried answer) need 13.7 at 25–75% baseline accuracy. The significant results survive stricter correction: under one BH correction over all 264 tests, every bold cell of Table 1 but one, and all but three ^f marks, stay significant. Finally, even greedy decoding is not fully repeatable: rerunning 1,200 identical plain-1.7B node-degree prompts changes correctness on 5.3% of them. We therefore compare net correct shares, and report the number of responses that switch outcome only descriptively.

Graph-blind solver. Besides the printed primer and the query targets, the solver sees the graph’s node and edge counts (§3), never the incident edge list. For edge count, it computes the answer from stated degrees, estimates it from mean clustering

with a fitted rule, or otherwise falls back to the majority answer of the fitting graphs. Its 16 exact rules, one heuristic and eight fitted rules are fit and scored on disjoint graphs. Fitted rules can fail outside their range: one RWSE rule infers density from return values to answer edge-existence questions, but scores below the majority answer at high density, which is why RWSE falls below no primer on that task in Table 2. The solver’s score therefore shows what one shortcut achieves, not an upper bound on what the full prompt conveys. Sparse graphs leave strong indirect clues (§4.1): from rounded clustering values alone, a separately fitted lookup answers 51.0% and 54.0% of degree queries at $p = .10$ and $p = .20$, against majority answers of 17.0% and 16.0%.

Primer	Mean chars	Degree	Neighbors	Edge count	Edge exists
none	0	14.4	0.0	3.0	73.5
degree	891	100.0	0.1	100.0	74.6
clustering	1,629	14.4	0.0	10.4	72.7
RWSE	2,949	21.0	0.1	3.0	55.4
all	4,691	100.0	0.8	100.0	74.1
components	37	14.4	0.0	3.0	73.5
filler	1,829	14.4	0.0	3.0	73.5

Table 2: Graph-blind solver accuracy (%) on the four variable-answer tasks, averaged over seven densities ($p = .10$ –.85, 40 nodes) and scored on separately generated test graphs, with mean primer length in characters (none adds no text). Components and filler score the same as no primer. The solver sees graph size and the query targets but no incident edges.

Scoring. The answer extractor parses 99.84% of responses. Neighbor-set exact match is the primary measure and set-F1 a secondary one: among the five contrasts whose exact match changes by at least 5 points, F1 moves the same way in three, at 0.11–0.17 times the size, and reverses in two. For edge existence, balanced accuracy is tested with paired permutation tests, BH-adjusted over the six conditions within each arm.

B Detailed Results

Table 3 gives intervals for selected contrasts; Table 5 reports every truncation contrast. Figures 3, 4 and 5 show degree effects across arms and densities; Figure 6 separates completion from accuracy on edge count. Other tables examine edge existence, cycle support and the plain-thinking gap.

Task	Arm	Primer	No primer	Δ	95% CI
Degree	1.7P	degree	60.50	+7.5	[+3.0, +11.8]
	1.7T	degree	76.25	+8.8	[+4.2, +13.2]
	4P	degree	99.25	-6.5	[-9.2, -4.0]
	4P	all	99.25	-8.0	[-11.0, -5.2]
Neighbors	1.7T	degree	78.75	+6.2	[+2.8, +10.0]
	4P	components	86.00	+9.2	[+6.0, +12.8]
	4P	filler	86.00	-12.5	[-17.2, -7.8]
Edge count	4P	degree	2.00	+27.2	[+22.8, +32.0]
Edge exists	1.7P	all	70.50	+11.5	[+6.2, +16.5]
	1.7P	filler	70.50	-15.8	[-20.0, -11.8]
	4P	clustering	90.50	+4.0	[+1.0, +7.0]
	4P	RWSE	90.50	-10.0	[-13.8, -6.2]
Node count	1.7P	clustering	31.50	+67.2	[+64.5, +70.0]
Cycle	1.7P	degree	90.75	-17.5 [†]	[-23.0, -12.2]
	1.7T	degree	97.00	-21.8 [†]	[-26.5, -17.5]

Table 3: Uncertainty for selected contrasts: no-primer accuracy (% of all prompts) and paired change (points) with its graph-bootstrap interval. Every unflagged contrast shown has BH $q < .05$; [†] marks at least 15% truncation on either side. Main-sweep point estimates for all four variable-answer tasks are in Table 1; all intervals are in the repository output `outputs/n40-sweep/primer_findings.txt`. Node count and cycle check have constant gold answers. P/T: plain/thinking.

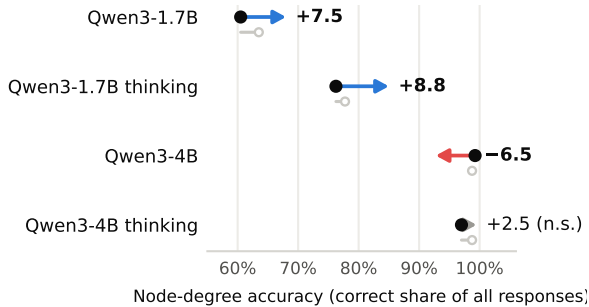


Figure 3: Node-degree accuracy per arm with no primer (dot) and with the degree primer (arrow), as in Table 1. Hollow: the clustering primer, which states no degree. Blue gains and red losses have $q < .05$ against no primer; gray does not.

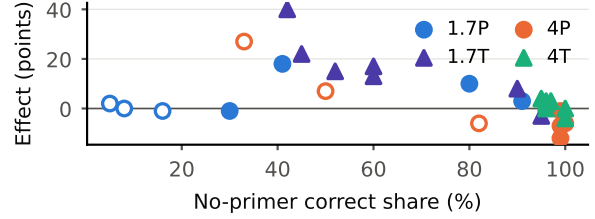


Figure 4: The degree primer's effect on node degree against the no-primer correct share, per arm and density. Hollow: plain arm at $p \geq .65$ (smaller budget).

Unaided accuracy	Answer stated	Side statistic
0-25%	+8.3 (16)	-1.7 (23)
25-50%	+16.1 (10)	+2.6 (20)
50-75%	+12.9 (8)	+3.9 (47)
75-90%	-5.8 (4)	+0.7 (51)
90-100%	-0.9 (28)	-0.9 (177)

Table 4: Mean accuracy change against no primer (points; number of cells in parentheses) by unaided-accuracy band, over unflagged arm-task-density cells of the four variable-answer tasks ($p \leq .50$, plus $p \geq .65$ for node degree and edge existence). *Answer stated*: degree and all on node degree and edge count. *Side statistic*: every other primer-task pair, including components and degree or all on neighbor sets and edge existence; filler is excluded.

Stated answers (§4.2). Answer-stating primers gain most, in points, where unaided accuracy is intermediate (Table 4; Figures 3 and 4). This holds when bands are formed on one half of the graphs and evaluated on the other, and at truncation thresholds from 5% to 50% (at least +11.6). On a log-odds scale, however, the gain is similar across the three lower bands, so the intermediate peak partly reflects the points scale; and the losses at high unaided accuracy come from plain 4B alone. Density, by contrast, has no detectable effect at a fixed unaided accuracy (in the main sweep, permutation $p = .28$ for answer-stating primers and $p = .85$ for side statistics). The one exception, a side-statistic effect on node degree that falls with density ($p = .006$), disappears once $p \geq .65$ is added ($p = .58$). Plain 4B spans both ends of Table 4: besides its gain at $p = .85$, the degree primer costs it 12.0 points at $p = .50$, where it answers 99.0% unaided (both $q < .01$; Figure 4). A printed answer must still be found and used. Across seven densities, even though the solver reads every degree correctly, accuracy under the degree primer ranges from 43% for plain 1.7B to 98% for thinking 4B. At high density, the degree primer does not detectably help plain 1.7B (+1.2 points, $q = .44$);

Task	Arm	none	degree	clustering	RWSE	all	components	filler
Degree	1.7P	0.00	+0.0	+0.0	+0.2	+0.0	+0.0	+0.0
	1.7T	0.75	+4.5	-0.5	+0.0	-0.5	-0.2	-0.5
	4P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4T	2.75	-2.5	-2.0	-0.5	-1.0	-0.5	-1.8
Neighbors	1.7P	0.00	+0.0	+0.0	+0.0	+0.2	+0.0	+0.0
	1.7T	1.25	+0.2	-1.0	-0.8	-0.5	-0.2	-0.8
	4P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4T	3.25	-0.2	+1.2	-0.2	+1.0	+0.2	-0.5
Edge count	1.7P	29.75	-20.2	-11.8	-11.8	-1.8	-2.5	-3.5
	1.7T	81.75	-0.8	+0.8	+5.0	-12.2	+0.5	+1.2
	4P	0.00	+1.8	+0.2	+0.5	+0.8	+0.8	+0.0
	4T	79.50	-20.8	-26.0	-12.2	-23.8	-25.0	-13.0
Edge exists	1.7P	0.00	+0.0	+0.0	+0.2	+0.0	+0.0	+0.0
	1.7T	0.25	+0.2	+0.0	+0.0	-0.2	+0.0	-0.2
	4P	0.00	+0.5	+0.0	+0.0	+0.2	+0.0	+0.0
	4T	0.25	+0.0	+0.8	+1.2	+1.0	+0.5	+0.0
Node count	1.7P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	1.7T	7.75	-5.0	-7.8	-7.8	-7.8	-0.5	-7.8
	4P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4T	0.00	+0.2	+0.2	+0.2	+0.0	+0.0	+0.0
Cycle	1.7P	9.00	+8.2	-8.2	-7.2	-8.0	-2.5	-2.8
	1.7T	3.00	+21.2	+0.5	-2.2	+0.8	+7.5	-3.0
	4P	0.75	+0.8	-0.2	+0.5	-0.2	+0.8	+0.5
	4T	0.75	+4.8	+0.0	-0.2	+0.5	+3.2	-0.5

Table 5: Complete main-sweep truncated shares for all six tasks: no-primer baseline (% of all prompts) and paired change (points). Together with Table 1, these give the *wrong* share: $wrong = 100 - correct - truncated$. P/T: plain/thinking; components and filler are controls. Truncation contrasts carry no significance marks.

instead, at $p = .85$ the model answers 39, the degree of a node joined to all others, on 70.8% of finished responses with the primer and 61.0% without. One source of error is a neighboring line: across seven densities, 17 of 57 wrong plain-4B lookups copy the degree of an adjacent sentence, more than chance predicts ($p = .017$). The combined primer is costlier still for dense plain 4B (§4.3; $q < .001$, against $q = .03$ for degree alone’s gain); Figure 5 shows this change for every arm and density.

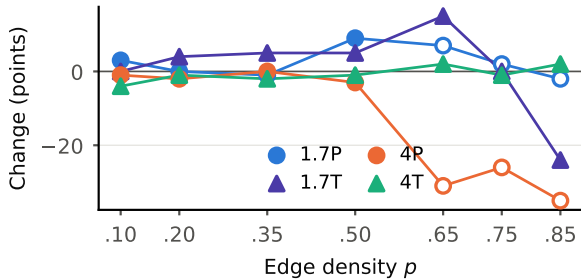


Figure 5: What adding clustering and RWSE to the same degrees changes on node degree (all three statistics minus the degree primer, points), per arm and density. Hollow as in Figure 4.

Controls and run-to-run noise (§4.3). Filler changes 6.2–18.6% of outcomes per arm, and the primers largely change the same questions. In every arm, every primer, components included, changes 38.5–69.1% of the questions filler changes but only 3.1–15.2% of the rest, and this gap holds within task and density ($q < .05$ for all twenty contrasts).

Part of this overlap is noise. When plain 1.7B’s node-degree prompts are generated a second time, 24 of 400 questions change outcome, and every condition changes these questions more often than the rest ($q \leq .018$). However, they account for only 11–16 of each condition’s 65–113 changes, and because each prompt was generated only twice, this undercounts how many changes noise alone causes. Filler’s changes also tend to be losses: where unaided accuracy is 25–75%, components, clustering and RWSE average +1.6 points, while filler averages -6.9 on the same arm-task-density cells. Across the four variable-answer tasks, the largest unflagged main-sweep gain without degrees is the component sentence’s 9.2-point increase in plain 4B’s *exact* neighbor-set accuracy; its set-F1 barely moves. The sentence is identical across all sampled graphs at $p \geq .20$, yet responses restate the queried node’s incident line much more often with it than with no primer or filler (98.5%, 63.0%, and 21.8%). Filler moves this response pattern in the opposite direction and lowers exact-match accuracy (Table 1). On plain 4B’s neighbor sets, restatements of the queried incident line are about equally accurate with the component sentence and without a primer (95% and 92%; 84% with filler), so the sentence’s gain comes mainly from restating more often.

Edge existence (§4.3). Primers change edge existence mainly in the plain arms (Table 6). There,

clustering and RWSE move plain 4B in opposite directions, and the combined primer helps plain 1.7B but harms plain 4B. Because the share of true edges changes with density, the table adds balanced accuracy, which weights a correct yes and a correct no equally; “yes share” is the fraction of finished responses answering yes.

Primer	1.7P			4P		
	Raw	Balanced	Yes	Raw	Balanced	Yes
none	70.50	79.86	56.2	90.50	93.22	35.7
degree	74.50	82.30	51.7	85.75	90.27	40.7
clustering	74.25	82.42	52.5	94.50	96.25	32.3
RWSE	75.50	82.68	50.1	80.50	86.69	46.3
all	82.00	87.71	44.8	80.50	86.69	46.1
components	62.25	74.23	64.5	87.75	91.64	39.0
filler	54.75	69.11	72.0	83.50	88.74	43.2

Table 6: Edge existence in plain arms (400 prompts per row): raw and balanced accuracy over all prompts and “yes” share among finished answers (%). Bold: balanced accuracy differs from no primer (paired permutation, BH $q < .05$ within arm). Thinking-arm balanced accuracy remains 97.61–99.83%; their full rows are in the repository output `outputs/n40-sweep/primer_findings.txt`.

False alarms rise with the number of shared neighbors (Table 7). In $G(n, p)$ that number is independent of whether the pair is joined, so this pattern suggests that models mistake proximity for adjacency. It is not a density artifact: within each density and without a primer, false alarms share on average 4.48 more neighbors than correct no’s for plain 1.7B ($p = .0005$) and 2.81 more for plain 4B ($p = .014$). Clustering changes stated reasons more than decisions: under clustering, plain 1.7B’s responses explicitly claiming a non-edge is an edge fall from 13.7% to 4.4%, but its false alarms fall only from 40.3% to 35.2%.

Plain arm	0	1	2–3	4+
1.7B	1.7	34.3	47.3	65.9
4B	0.2	4.6	19.0	38.8

Table 7: False “yes” on non-edges (% of finished answers), by number of shared neighbors in the true graph; all seven conditions pooled (including no primer and filler) at $p \leq .50$. Bin sizes (1.7B / 4B): 524/524, 350/350, 448/447, 728/727. Pooling across densities is descriptive.

Dense graphs (§4.3). Clustering raises dense plain 4B’s node-degree accuracy although its printed values barely vary there: at $p = .65$, a graph’s clustering values have SD 0.021, and rounded RWSE takes 2.08 distinct value pairs per

graph against 12.04 distinct degrees. Neither added length nor a change of strategy explains the gain: length-matched filler does not help ($+2.0$, $q = .59$), and neighbor enumeration is about as common with clustering as without (69% and 74%). The gain may therefore come from how the added numeric sentences change the prompt, which this design cannot isolate.

Against the thinking mode (§4.4). Plain-thinking comparisons use the same graphs; we call the modes equivalent only when the 90% interval for their difference lies within ± 3 points. The one equivalence in Table 8 comes from the component sentence: on neighbor sets, the 90% interval of plain 4B’s gap to thinking 4B without a primer is $[-2.0, +2.5]$, within the prespecified ± 3 -point margin. For plain 1.7B’s neighbor sets under degree, the gap is also nonsignificant ($q = .24$), but its interval reaches $+6.2$, so equivalence is not established. To avoid selecting a primer on the graphs it is scored on, each row’s best primer is chosen on even-index graphs and scored on odd-index graphs, and vice versa; both halves chose the same primer.

Task	Model	No primer gap	Best primer for plain	Gap with best primer	Gap with CI/RWSE
Degree	1.7B	+15.8	all	+5.5	+11.8
	4B	−2.2	components	−2.5	−1.8
Neighbors	1.7B	+6.2	degree	+2.8	+5.0
	4B	+9.5	components	+0.2^e	+8.5
Edge count [†]	1.7B	+14.5	degree	+11.5	+15.0
	4B	+18.0	degree	−9.2	+19.2
Edge exists	1.7B	+28.5	all	+17.0	+23.5
	4B	+9.2	clustering	+5.2	+5.2

Table 8: Gap in correct share (points): the same model thinking without a primer minus plain under no primer, the plain arm’s best primer selected on held-out graphs (see text), or the better of clustering and RWSE. Negative values put plain ahead. Bold: paired BH $q < .05$; [†]: at least 15% truncation on one side (Table 5), so no significance marks. Edge-existence gaps are in raw accuracy; balanced accuracy is in Table 6. ^e: the 90% interval lies within the prespecified ± 3 -point equivalence margin.

Thinking mode (§4.4). Under the degree primer, thinking 1.7B still lists the neighbors in 90% of responses at $p \geq .35$, and thinking 4B ends on the stated value in 93% of finished conflict responses, more often than thinking 1.7B. Because thinking responses are long, truncation drives many of the changes in these arms: of the answers a condition changes, either response hit the token budget in 44.4–64.5% for thinking 1.7B and 83.3–95.2% for thinking 4B (Figure 6 shows this for thinking 4B’s

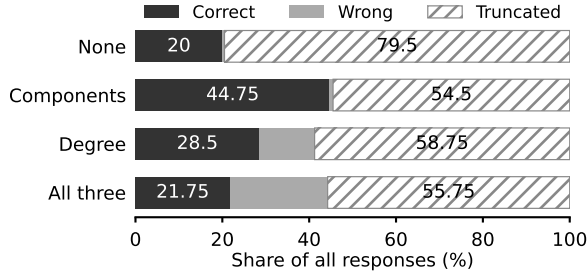


Figure 6: Thinking 4B on edge count: correct, wrong and truncated shares of *all* responses.

edge count). The degree primer puts plain 4B ahead of thinking 4B on edge count, but the comparison mostly reflects completion: thinking 4B is correct on 97.6% of the edge-count responses it finishes, while 79.50% of its responses exhaust the token budget.

Graph support (§4.5). Table 9 gives the values behind that section: no clustering or RWSE change in joint correctness is significant ($q \geq .16$). The table also reports valid cycle support.

Primer	Joint correct		Valid cycle support	
	1.7P	4P	1.7P	4P
none	54.00	85.75	34.2	58.5
degree	62.50	76.50	50.2	76.5
clustering	58.00	86.25	40.2	54.2
RWSE	58.25	83.75	32.0	33.2
all	61.75	77.50	38.8	45.2
components	54.50	95.00	29.8	63.7
filler	52.00	72.50	31.2	45.8

Table 9: Two graph-support checks in the plain arms (% of all prompts). *Joint correct*: both the degree and the exact neighbor set are right for the same queried node. *Valid cycle support*: a correct, finished cycle “yes” backed by a valid argument, a real cycle or the edge-count argument. Bold: BH $q < .05$ against no primer. Response categories are in Table 10.

Table 10 breaks every correct, finished cycle-check answer down by the argument it cites, and many rest on invented cycles. For example, on one $p = .20$ graph plain 1.7B without a primer answers with $0 \rightarrow 4 \rightarrow 1 \rightarrow 5 \rightarrow 14 \rightarrow 13 \rightarrow 15 \rightarrow 16 \rightarrow 18 \rightarrow \dots \rightarrow 38 \rightarrow 39 \rightarrow 0$, which climbs the ids almost in order; steps such as $15 \rightarrow 16$, $21 \rightarrow 25$ and $38 \rightarrow 39$ are not edges. The degree primer raises valid support for plain 4B as well as plain 1.7B (§4.5), while RWSE, all three statistics and filler reduce plain 4B’s. Thinking arms cite real cycles far more often (79.9% and 96.5% of correct answers without a primer), but RWSE turns 42.5% of thinking 4B’s correct answers into walks

Arm	Primer	Correct	Cycle named			No cycle	
		“yes” (<i>n</i>)	Real	Invented	Not a cycle	Total	Edge count
1.7P	none	363	36.9	43.5	16.0	3.6	0.8
	degree	293	13.0	20.5	1.4	65.2	55.6
	clustering	397	39.5	48.1	7.8	4.5	1.0
	RWSE	362	33.7	40.3	19.9	6.1	1.7
	all	391	38.4	50.4	7.9	3.3	1.3
	components	371	27.8	40.4	25.3	6.5	4.3
	filler	375	33.3	55.5	9.6	1.6	0.0
1.7T	none	388	79.9	17.8	1.0	1.3	0.5
	degree	301	61.5	29.6	0.0	9.0	9.0
	clustering	386	80.6	19.2	0.0	0.3	0.3
	RWSE	397	68.0	20.7	10.6	0.8	0.5
	all	385	73.0	21.8	3.1	2.1	1.8
	components	358	74.9	22.6	0.0	2.5	2.0
	filler	400	77.2	19.8	2.5	0.5	0.0
4P	none	396	34.1	23.0	6.8	36.1	25.0
	degree	387	13.4	3.9	2.1	80.6	65.6
	clustering	398	48.7	31.2	1.0	19.1	5.8
	RWSE	395	33.2	20.8	38.7	7.3	0.5
	all	395	45.1	33.4	6.6	14.9	0.8
	components	388	39.2	18.6	3.4	38.9	26.5
	filler	394	44.2	28.2	5.3	22.3	2.3
4T	none	397	96.5	3.3	0.0	0.3	0.0
	degree	375	90.9	5.1	0.0	4.0	2.7
	clustering	397	95.5	4.5	0.0	0.0	0.0
	RWSE	398	55.0	2.5	42.5	0.0	0.0
	all	395	92.7	6.3	0.0	1.0	1.0
	components	384	92.2	4.4	0.0	3.4	2.9
	filler	399	94.2	5.0	0.5	0.3	0.0

Table 10: What the correct cycle-check answers rest on (main sweep; every correct answer is “yes”). *Correct (n)*: count of correct, finished answers; the other columns are % of those *n*, and Real, Invented, Not a cycle and Total sum to 100. *Real*: every step of the cycle is an edge; *Invented*: a step is not an edge; *Not a cycle*: a length-2 or retraced walk. *Edge count*: the answers under Total that argue from the number of edges, identified by their wording. The valid arguments are Real and Edge count, which is valid because 40 nodes with at least 40 edges must contain a cycle. Bold (Real, Invented only): differs from the same model without a primer (paired, BH $q < .05$).

that are not cycles. Finally, truncation explains almost all of the degree primer’s cycle-check loss for thinking 1.7B, but less than half for plain 1.7B (Tables 3 and 5).

Node count. This task cannot show graph reading, because every graph has 40 nodes, all named in the encoding’s first line. Plain 1.7B often mistakes the last identifier, 39, for the count, and most primers reduce that error: clustering nearly eliminates it (68.5% to 1.2% of finished answers), and even filler, which states nothing, raises accuracy by 39.8 points. The large effect in Table 3 is therefore not specific to clustering and does not show graph reading.

C Pilot and the Clustering Screen

Pilot. On the published GraphQA graphs (5–19 nodes), Gemma 4 E4B/12B (Gemma Team, 2026) and Qwen3-8B/14B, in both modes, were near ceiling without a primer on most tasks. Edge count remained difficult. These pilot results motivated the 40-node graphs and smaller models; they are not pooled with the main sweep.

The clustering effect. Because clustering’s gain on plain 1.7B’s node degree emerged from a 40-cell screen, Table 11 retests it. The unscreened, fresh-seed, fixed-mean-degree and Qwen3-8B rows use graphs the screen did not see; the $p \leq .50$, against-filler and $p \geq .65$ rows use the full 400-graph-per-density runs, which include the screened graphs. The screened-graph row is descriptive only, because it reuses graphs from the initial search. The gain holds up on unscreened graphs, a fresh seed and the fixed-mean-degree grid, where it is largest in the densest cell (20 nodes, $p = .842$: +16.5) but reverses in the other 20-node cell ($p = .421$). However, it is not detected on 40-node graphs at $p \geq .65$, nor in Qwen3-8B, which is near ceiling in most cells (Table 11). Thinking 1.7B shows a similar split in the follow-ups: clustering gains it 2.5 points at $p \leq .50$ ($q = .043$) but loses 5.4 at $p \geq .65$ ($q = .0015$), where filler loses 11.4.

Test	N	Gain	Statistic	Correction
Screened graphs	700	+1.29	$p = .47$	none
Unscreened graphs	2,100	+2.1	$p = .031$	none
$p \leq .50$	1,600	+3.8	$q = .0035$	BH
Fresh seed	1,600	+4.2	$p = .00055$	Bonf. = .022
Against filler	1,600	+3.2	$q = .014$	BH
Fixed mean degree	3,200	+6.2	$q < .001$	one test
$p \geq .65$	1,200	−0.7	$q = .57$	BH
Qwen3-8B	3,200	−1.0	$q = .05$	one test

Table 11: Node-degree accuracy change (points) for plain Qwen3-1.7B, except the Qwen3-8B row. N is paired prompts. Comparisons use no primer unless labeled against filler. *Screened*: the 700 graphs the discovery screen used; *unscreened*: the other 2,100; *fresh seed*: newly generated graphs; *fixed mean degree*: the 20–160-node grid. p is unadjusted; q is BH-adjusted (equal to p in a one-test family); the fresh-seed Bonferroni value corrects across the 40 screened cells.

How clustering helps is unclear. No sampled response explicitly invokes clustering, and the gain does not vary detectably with the queried node’s coefficient ($p = .50$), though it grows with the node’s degree ($p = .012$). The clearest pattern is list position: the gain is +5.01, +0.75 and −0.11 points for queried nodes in the first, middle and

last list thirds, and, adjusting for degree, clustering and density, it shrinks by 1.74 points per ten later positions ($p = .018$), while filler and component controls show no corresponding gradient. These patterns motivate a position-based explanation but do not prove one: clustering resolves some prior copying errors while introducing others, and components and filler also reduce missed neighbors, so no clustering-specific mechanism is isolated. Shuffling node order would test the position hypothesis directly.